



Mathematics Presentation

July 22, 2026

Outline

- 1 Definition & Forms
- 2 Graphical Properties
- 3 The Discriminant & Three Cases
- 4 Applications & Optimization

What is a Quadratic Function?

Definition

A **quadratic function** is a second-degree polynomial function in one variable. The highest exponent of the variable is 2.

Three Main Algebraic Representations

- 1 **Standard Form:** $f(x) = ax^2 + bx + c$ ($a \neq 0$)
- 2 **Vertex Form:** $f(x) = a(x - h)^2 + k$
- 3 **Factored (Intercept) Form:** $f(x) = a(x - r_1)(x - r_2)$

Forms of Quadratic Functions

Form	Equation	Key Insight Provided
Standard	$f(x) = ax^2 + bx + c$	y-intercept is $(0, c)$
Vertex	$f(x) = a(x - h)^2 + k$	Vertex is (h, k)
Factored	$f(x) = a(x - r_1)(x - r_2)$	x-intercepts are r_1, r_2

Table: Comparison of Quadratic Forms

Anatomy of a Parabola

The graph of a quadratic function is a U-shaped curve called a **parabola**.

Key Features

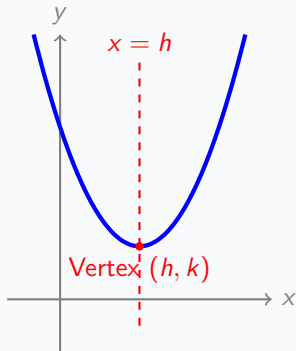
- **Orientation:**

- $a > 0 \implies$ opens **upward** ($\cup$)
- $a < 0 \implies$ opens **downward** ($\cap$)

- **Vertex:** Point of maximum or minimum value, located at (h, k) where:

$$h = -\frac{b}{2a}, \quad k = f(h)$$

- **Axis of Symmetry:** Vertical line $x = -\frac{b}{2a}$



The Discriminant (Δ)

For the quadratic equation $ax^2 + bx + c = 0$, roots are given by:

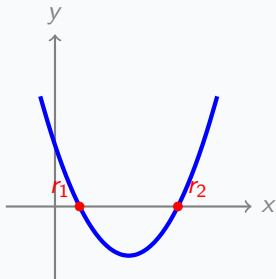
$$x = \frac{-b \pm \sqrt{\Delta}}{2a} \quad \text{where} \quad \Delta = b^2 - 4ac$$

The Three Discriminant Cases

- **Case 1 ($\Delta > 0$):** Two distinct real roots. The parabola crosses the x-axis twice.
- **Case 2 ($\Delta = 0$):** One repeated real root. The vertex touches the x-axis at exactly one point.
- **Case 3 ($\Delta < 0$):** Two complex conjugate roots. The parabola does not intersect the x-axis.

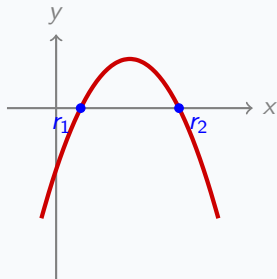
Case 1: $\Delta > 0$ (Two Distinct Real Roots)

Opening Upward ($a > 0$)



Intersects x-axis twice ($y_{\text{vertex}} < 0$).

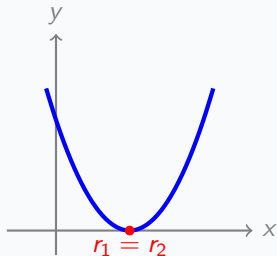
Opening Downward ($a < 0$)



Intersects x-axis twice ($y_{\text{vertex}} > 0$).

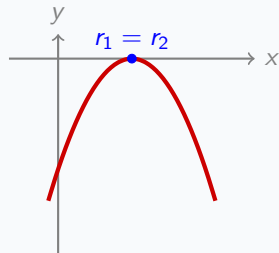
Case 2: $\Delta = 0$ (One Repeated Real Root)

Opening Upward ($a > 0$)



Vertex touches x-axis ($y_{\text{vertex}} = 0$).

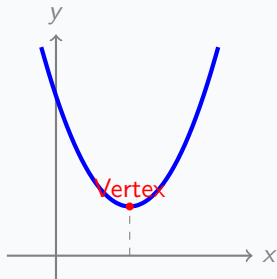
Opening Downward ($a < 0$)



Vertex touches x-axis ($y_{\text{vertex}} = 0$).

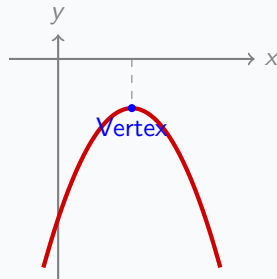
Case 3: $\Delta < 0$ (Two Complex Conjugate Roots)

Opening Upward ($a > 0$)



Lies strictly above x -axis ($y > 0$ everywhere).

Opening Downward ($a < 0$)



Lies strictly below x -axis ($y < 0$ everywhere).

Real-World Applications

Projectile Motion

Height of a projectile over time t :

$$h(t) = -\frac{1}{2}gt^2 + v_0t + h_0$$

- Max height occurs at vertex $t = \frac{v_0}{g}$.
- Flight time found when $h(t) = 0$.

Economics & Optimization

Profit function $P(x) = R(x) - C(x)$:

$$P(x) = -ax^2 + bx - c$$

- Maximum profit achieved at $x = -\frac{b}{2a}$.
- Break-even points found at $P(x) = 0$.

Summary & Key Takeaways

Essential Reminders

- ① $\Delta > 0$: Graph crosses x -axis twice (2 real roots).
- ② $\Delta = 0$: Graph touches x -axis at 1 vertex point (1 repeated root).
- ③ $\Delta < 0$: Graph never touches x -axis (2 complex roots).
- ④ Orientation depends entirely on sign of a : $a > 0$ ($\cup$) vs. $a < 0$ ($\cap$).

Questions & Discussion